STM32H743 High-Performance MCU - DATASHEET

Manufacturer: STMicroelectronics
Part Number: STM32H743ZIT6
Category: Microcontroller / ARM Cortex-M7 MCU
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Overview

The STM32H743 is a high-performance microcontroller based on the ARM Cortex-M7 core running at up to 480 MHz with double-precision FPU. Features 2 MB Flash, 1 MB SRAM, hardware JPEG codec, Chrom-ART Accelerator for graphics, and a rich set of peripherals including Ethernet MAC, dual USB OTG, FDCAN, and SDMMC. L1 cache: 16 KB I-cache + 16 KB D-cache.

Analog Peripherals

ADC: 3× 16-bit ADCs with up to 3.6 MSPS in interleaved mode. DAC: 2× 12-bit DACs. Comparators: 2× ultra-low-power comparators. Op-amps: 2× operational amplifiers. Temperature sensor: integrated. VBAT monitoring channel. ADC features: hardware oversampling (up to 1024x), differential input mode, DMA support.

Key Specifications Summary

core: CortexM7. freq: 480MHz. flash: 2MB. sram: 1MB. adc: 3x16bit. eth: yes. usb: 2xOTG. can: FDCAN.